



SPACE — FULL MIND MAP

अंतरिक्ष · Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this chapter:

1. Indian Space Beginnings, Space Pioneers & Shuttle Milestones
2. Planetary Missions, Gravitational Waves, Black Holes, Stars & Solar Storms
3. Geostationary Satellites, Space Telescopes & Observatories
4. ISRO, INSAT, Communication & Remote Sensing
5. PSLV Missions, Navigation Systems & Launch-Vehicle Evolution
6. ISRO Centres, Sriharikota, Thumba & Aviation Basics
7. Moon Missions: Chandrayaan, LUPEX, Luna & Aditya-L1
8. Mangalyaan, China & International Space Cooperation
9. NASA Missions, Planets, Jet/Rocket Principles & Cryogenics
10. Indian Launchers, Satellites, EDUSAT, GSAT & BHUVAN

1. Indian Space Beginnings, Space Pioneers & Shuttle Milestones

- Dr. Vikram Ambalal Sarabhai ek Indian Physicist aur astronomer the jinhone India me space research ki shuruaat ki thi. Unhe Indian space programme ka janak maana jata hai. Dr. Vikram Sarabhai ko India ke space programme ka janak maana jata hai. Aryabhata India dwara launch kiya gaya pehla satellite tha. Iska naam isi naam ke mahaan Indian astronomer ke naam par rakha gaya tha. [Q1–Q3]
- India ne apne pehle satellite 'Aryabhata' ke launch ke saath space age me pravesch kiya. Aryabhata ko April 19th, 1975 ko successfully space me launch kiya gaya tha. Prof. U.R. Rao wo scientist the jinhone India ke pehle space satellite 'Aryabhata' ko design kiya tha. Rao ne 1972 me India me satellite technology ki establishment ki responsibility li thi. [Q4–Q6]
- Laika ek Soviet space dog tha jo space me Earth ko orbit karne wala pehla animal bana gaya. 3 November, 1957 ko 'Laika' naam ke ek dog ko Sputnik-2 spacecraft ke through space me bheja gaya tha. 12 April, 1961 ko astronaut Yuri Gagarin Russian Vostok-1 spacecraft se space me jane wale pehle human bane. [Q7–Q9]
- Valentina Tereshkova (Russian astronaut) space me jane wali pehli mahila bani, jab unhe 16 June, 1963 ko Vostok-6 mission par launch kiya gaya tha. Apollo-11 wo space flight thi jisme 20 July, 1969 ko Moon par do logon ke saath lunar module 'Eagle' ko utara tha. Oleg Kononenko ek Russian cosmonaut hain. unhone International Space Station (ISS) par 5 long-duration missions ke dauran space me 1110.6 din total kiye hain. [Q10–Q12]
- Currently, NASA ke astronaut Francisco (Frank) Rubio ke naam 370.9 dinon ke saath kisi American dwara sabse longest single spaceflight ka record hai. Ve 27 September, 2023 ko Earth par laute. Tim Peake British nationality ke European Space Agency (ESA) ke astronaut aur ek former International Space Station (ISS) crew ke member hain. Atlantis ek Space Shuttle hai (Space Station nahi). Columbia, Challenger, Discovery aur Atlantis sabhi NASA dwara operated Space Shuttle the. [Q13–Q15]
- Curiosity (Curiosity): Mars ko explore karne wala rover (A-3). MESSENGER (MESSENGER): Mercury ka study karne ke liye launch kiya gaya NASA ka spacecraft (B-4). Atlantis NASA ka last aur aakhiri Space Shuttle tha. Ise 8 July 2011 ko launch kiya gaya tha aur 21 July 2011 ko ye utara, jisse Space Shuttle programme ka end ho gaya. [Q16–Q17]

⚠ EXAM TRAP — Q13 — Spaceflight Record is Updated/Time-Sensitive

► Source itself Q13 ko updated/disputed mark karta hai. Longest single NASA spaceflight aur cumulative-space-time records date ke saath change ho sakte hain; exam-year context follow karein.

2. Planetary Missions, Gravitational Waves, Black Holes, Stars & Solar Storms

- 'VERITAS' aur 'DAVINCI+' 1990 ke decade ke baad se Venus ke liye NASA ke pehle mission honge. Cassini-Huygens (Cassini-Huygens): Saturn aur uske rings ka study (A-2). Cassini-Huygens (Cassini-Huygens) Saturn planet (Saturn) par bheja gaya ek human rahita spacecraft tha, na ki Venus par. Isliye yugma 1 sahi matched nahi hai. [Q18–Q20]
- NASA ka Cassini spacecraft Saturn planet (Saturn) aur uske rings aur Moonon ki complex system ka detect lagaane ke liye design kiya gaya ek mission tha. Cassini chhalledaar planet Saturn ka study karne ke liye NASA dwara developed ek spacecraft tha. Ye prayog involvd LASER interapherometar space enteena (eLISA) ko refer karta hai, jo gravitational waves (gravitational waves) ka detect lagaane aur accurate roop se measure karne ke liye design kiya gaya ek space jaanch hai. [Q21–Q23]
- ELISA project ka primary purpose space se To detect gravitational waves aur unakaa nireekshan karna hai. 2009 me, British scientists ne ghoshana ki ki unhone 'greivitee traiktara' naam ka ek vishesh spacecraft design kiya hai jo ek Asteroid ko rokakar aur uske prakshepavakr ko badalakar Earth ko vinaashakaaree takaraav se bachaa sakta hai. Scientists ne do black hole ke vilaya ko dekhaa, jisse spesataaim ke taane-baane me heavy amount me energy utpann hui. Is observation ka importance ye hai ki pehli baar LIGO observatoryon dwara gravitational waves (gravitational waves) ka directly detect lagaay gaya. [Q24–Q26]
- Gravitational waves ka pehla pratyaksh observation 14 September, 2015 ko kiya gaya tha aur 11 pharavaree, 2016 ko LIGO dwara iski ghoshana ki gayi thi. Space me maaikroev (kosmika maaikroev baikagraaunda) ka detect lagaana aur redashipt ghatana (sudoora galaxyon se light ka spektram ke laala chhora par sthanaantarit honaa) ka observation scientists dwara brahmaand ke niranter expansion ke primary saakshy ke roop me uddhrit kiya gaya hai. Vistaarit brahmaand ki concept Doppler effect par based hai. Edavin habala ne sudoor galaxyon se light ke redashipt ko dekhakar ye determine karne ke liye Doppler effect ka use kiya ki brahmaand ka expansion ho rahaa hai. [Q27–Q29]

- Aainsteen ke general relativity ke principle ki bhavishyavaanee hai ki ek heavy shareer space-time ko modata (warp) hai (statement 3). Choonki gravity dwara space-time vakra hota hai, isliye isme se gujarane wala light bhi mudaa/prabhaavita hota hai (statement 1). General relativity ke sameekaranon ne naturally ek speedsheela, vistaarit (ya sikudate) brahmaand ko bhi nihita kiya (statement 2). Isliye sabhi sahi hain. 'iventa horaaijana' (Event Horizon) aur 'singulairitee' word black hole se jude hain. 'stringa thyoore' aur 'staindarda model' maulik physics concepten hain jo particles ki baatacheet ki vyakhy karti hain. Ye sabhi brahmaand ke observation aur samajh se related hain. Black hole vaastav me a collapsed level hai. Stellar mass wale black hole taba banate hain jab bahut bade stars apne jeevan chakra ke end me apne hi gravity ke under dhaha jate hain. [Q30–Q32]
- Black hole ek aisee vastu hai jo itanee saghan hoti hai ki iska gravity bala light ya kisi other cheeja ko baahar nikalane se rokane ke liye kaaphee majaboot hota hai. Choonki koi light baahar nahi nikalata hai, isliye ise directly telescope se nahi dekha ja sakta hai. Black hole apne high gravity ki wajah se usa par padane wale sabhi radiationon ko avashoshit (absorb) kar leta hai. Yahaan tak ki light bhi isase bacha nahi sakataa. Comet (Comets) Sun ke charon direction apni highly lambee, Elliptical orbiton ke liye jane jate hain. Ve Sun se bahut doora ek important time vyateet karte hain aur phira fastee se iske paas aate hain. [Q33–Q35]
- Sun se radiation aur saura air dhoola aur gas ke particles ko Comet se doora dhakelatee hai. Isliye, Comet ki poonchh hamesha Away from the Sun ki direction (Away from the Sun) hoti hai. Sun ki chamakeelee external layer ko Photosphere ke roop me jana jata hai. Photosphere word ka meaning hai 'prakaasha ka golaa' - jo upayukt hai kyonki ye wo layer hai jo sabse zyada light emit karti hai jise hama Earth se dekhate hain. Neutron stars huge suparajaayant stars ke dhaha gaye core hain. Ve highly saghan (dense) hote hain. [Q36–Q38]
- Pulsar (pulsars) highly magnetic, fastee se Rotating neutron stars hain jo apne magnetic dhruvon se electromagnetic radiation ke beema emit karte hain. Sephids ve stars hain jo time-time par chamakate aur manda hote hain. (jodaa 1 incorrect hai). Huge stars (Giant stars) me baune stars ke comparison me zyada garma aur saghan core hote hain. Isliye, unamen nuclear pratikriyaon ki dara zyada hoti hai (statement-II sahi hai). [Q39–Q41]
- Earth ka magnetic field ek security kavach ki tarah kaam karta hai. Ye saura air (solar wind) se electrical charged particles ko dhruvon ki direction moda deta hai, jisse ve Earth ki surface tak nahi access paate hain. Saura jvaalaaen (Solar flares) Earth par electrical chumbakeey vyavadhaan paidaa kar sakti hain. Ve jeepeees ko baadhit kar sakte hain, paavar grida ko damage pahuncha sakate hain, satellite orbiton ko pareshaan kar sakte hain, shortavev rediyo communication ko baadhit kar sakte hain aur teevra aurora paidaa kar sakte hain. Earth ke vaayumandal me punah praveshe ke dauran, ek spacecraft highly air gharshan ka anubhav karta hai. Ye highly garmee aasapaas ki air ko aayanit karti hai, jisse shuttle ke charon direction ek Plasma liphaapha bana jata hai. [Q42–Q44]

3. Geostationary Satellites, Space Telescopes & Observatories

- Statement 1 incorrect hai kyonki ek geostationary satellite bhoomadhy line se lagbhag 35,786 kilomeetar (lagbhag 36,000 km) oopara space me located hota hai, 10,000 km nahi. Geostationary orbit (geostationary orbit) bhoosamakaalik (geosynchronous) orbit ka ek specific type hai (1). Ye poori tarah se golaakaar (circular) honee chaahie (2) aur theeka Earth ki bhoomadhy line ke tala me located honee chaahie (3). ek geostationary (geostatic) sthiti taba mila hoti hai jab koi vastu (jaise A Satellite) Earth ki usee ghoornee speed (rotational speed) se Earth ki orbit karti hai. [Q45–Q47]
- Orbit ki duration ko Earth ke ghoornan (lagbhag 24 hours) se accurate roop se mela khaane ke liye, ek bhoos-thaitika satellite ko bhoomadhy line se lagbhag 35,786 kilomeetar ki ek bahut hi vishishta, nishchit elevation par orbit karanee chaahie. Exosphere Earth ke vaayumandal ki sabse external layer hai, jo external space ke shoony me lupta ho jati hai. Communication Satellite ~36,000 km ki elevation par orbit karte hain, jisse ve majabootee se Exosphere ke andar established ho jate hain. Earth ke bhoomadhy line par ek hi point par poori tarah se sthira rahane ke liye, ek geostationary satellite ko apni ek orbit theeka usee time me poori karanee chaahie jitana time Earth ko apni dhuree par ek baara revolving me lagata hai, jo ki 24 hours hai. [Q48–Q50]
- Jaisaa ki establish kiya gaya hai, ek geostationary satellite 24 hours me ek orbit poori karta hai, jo Earth ke ghoornan ke saath poori tarah se mela khaata hai. Earth ki surface (bhoomadhy line par) par ek point ke saath poori tarah se sinkranaaij (synchronized) rahane ke liye, geostationary satellite ka patha vrittaakaar (circular) orbit honaa chaahie. ek andaakaar (elliptical) orbit ki wajah se satellite ki speed me bhinnata hogee, jisse tulyakaalan (synchronization) toota jaaegaa. Kepalar space telescope 2009 me NASA dwara launch ki gayi ek specially design ki gayi space observatory thi. Iska specific mission milkee ve (galaxy) ke hamare field ka sarvekshan karna tha taaki rahane yogya field me ya uske aasa-paas (Earth-like Planets) saikadon Earth ke shape aur chhote planeton ki discover ki go sake. [Q51–Q53]
- 1999 me launch ki gayi NASA ki major chandra eksa-re observatory ka naam shaanadaar Indian-American Nobel Prize vijeta aur khagol scientist subrahmanyam chandrashekhar ke sammaan me rakha gaya tha, jinhen huge sistars ke development par unke kaam ke liye jana jata hai. Statement 1 incorrect hai kyonki NASA (USA) aur Russia ke alawa, European Union aur Japan ki space ejensiyon ne bhi India se pehle space me isee tarah ki suvidhaan (observatory) launch ki hain. Antaarkatik barpha me depression me dabee IceCube (IceCube nyootrino observatory), specific roop se world ki sabse badi nyootrino telescope ke roop me classified hai, jisme ek ghana kilomeetar barpha shamil hai. [Q54–Q56]
- Jab ye question poochha gaya tha (2002), to haanale, laddaakh (4,500 meetar ki elevation) me located Indian astronomical observatory (Indian Astronomical Observatory) world ka sabse ooncha optical telescope tha. Isliye die gaye options me India correct answer hai. Nota: aaja, chilee me tokyo worldvidyalay ki ataakaama observatory (5,640 meetara) ke paas full hightam record hai. [Q57–Q58]



EXAM TRAP — Largest / Highest Observatory Facts

► 'World's biggest telescope' ya 'highest observatory' jaise superlative facts technology ke saath change ho sakte hain. Source-year wording ko current record samajhkar blindly memorize na karein.

4. ISRO, INSAT, Communication & Remote Sensing

- ISRO India ki primary space Agency hai. Iska full form Indian Space Research Organisation hai, jiski establishment 1969 me National development ke liye space technology ka use karne ke liye ki gayi thi. INSAT-3B ko 21 march 2000 ko launch kiya gaya tha. Is duration ke dauran, Dr. Krishnaswami Kasturirangan (K. Kasturirangan) Indian space research organisation ke major the (unhone 1994 se 2003 tak adhyaksh ke roop me work kiya). Usa duration ke dauran jab ye question originally poochha gaya tha (2008), do jee maadhavan naayar ISRO ke adhyaksh the. [Q59–Q61]
- Jab ye question 2018 me poochha gaya tha, taba K. Sivan ne padabhaar sanbhaala tha. Haalaanki, vartamaan parivisible (2025) ko dekhate hue, do vee naaraavehiclen ISRO ke adhyaksh hain. Isliye, 'None of the above' sabse safe correct answer hai. Janavaree 2022 me, prakhyat rocket scientist do. S. Somanath ko K. Sivan ke sthana par Indian space research organisation (ISRO) ke dasaven adhyaksh ke roop me niyukt kiya gaya tha. Jab 2024 ki shuruaat me ye question poochha gaya tha, taba S. Somanath adhyaksh the. Haalaanki, janavaree 2025 tak, do vee naaraavehiclen ne ISRO ke vartamaan adhyaksh ke roop me padabhaar planetn kar liya hai. Isliye, koi bhi option full vartamaan vaastavikata ko nahi show karta hai. [Q62–Q64]
- Vyapak kavarej ke liye India vaastav me apne telecom network ko satelliteon (jaise INSAT aur GSAT shrinkhalaa) se connect karta hai. To, abhistatement (A) true hai. Indiyen neshanal satellite system (INSAT) telecom, prasaarana, weather science aur discovery aur bachaav functions ko poora karne ke liye ISRO dwara launch kiye gaye bahuuddesheey geostationary (geostationary) satelliteon ki ek series hai. ek bahupurposeeey system ke roop me, INSAT shrinkhala apni telecom bhoomikaaon ke alaava ek Meteorological Satellite network (important Weather forecast data provide karne) ke roop me heavy roop se kaam karti hai. [Q65–Q67]
- Modern geostationary Communication Satellite jaise INSAT aur GSAT data ki badi amount ko sanbhaalane ke liye C-band, Ku-band aur Ka-band jaise high-frequency maaikroev band ka use karte hain. Simplex communication ek communication chainal hai jo Only one dishaa me information send karta hai. Choonki India ke paas us time enough powershaalee rocket nahi tha, isliye iska pehla dedicated Earth observation satellite, aaeaaaraesa-1e (IRS-1A), march 1988 me Soviet sangha ke Baikanour kosmodrom (vartamaana kajaakistaan me) se dhruveey Sun-tulyakaalika orbit me launch kiya gaya tha. [Q68–Q70]
- Sudoor sanvedan (Remote sensing) satellite Earth ki surface ki detailed tasveeren lete hain. Ye data phasal ke svaasthy ka assessment karne (1), bhooyal ke liye bhaugolik maarkar discoveryne (2), aur khanij bhandaar ka detect lagaane (3) ke liye amooly hai. Haalaanki, unhe teevee/phone signal transmit karne ke liye design nahi kiya gaya hai (telecom - 4) aur na hi ve sadaka-level ki laaiva kaara speedvidhiyon (yataayata study - 5) ki nigaraanee kar sakte hain. Isliye 1, 2 and 3 only sahi hain. [Q71–Q72]

⚠ EXAM TRAP — Q61, Q62 & Q64 — ISRO Chairman Questions are Outdated

► Source in questions ko outdated mark karta hai. ISRO Chairman ka answer exam ke notification/year ke according update karna chahiye; historical chairman questions ko unke stated year se padhein.

5. PSLV Missions, Navigation Systems & Launch-Vehicle Evolution

- Isase pehle ki India bade Communication Satelliteon ko high geostationary orbiton me dhakelane me saksham apne svayan ke hevee-liphta rocket (jaise GSLV) developed karta, wo mainly French Guana me Kourou se launch kiye gaye European eriyen rocketon par depend tha. INSAT-3C ko 2002 me Kourou se launch kiya gaya tha. INSAT-4 satellite series me pehla, INSAT-4A, India me DTH television prasaaran ko badhaava dene ke liye 22 disanbara, 2005 ko Kourou, French Guana se ISRO dwara successfully launch kiya gaya tha. ek aitihaasik upalabdhi me, ISRO ke PSLV-C37 ne 15 pharavaree, 2017 ko ek hi launch ke dauran 104 satelliteon (international graahakon ke mainly chhote nano satelliteon) ko successfully orbit me establish kiya. 2021 me spesaeaks ke isase aage nikalane tak ye world record rahaa. [Q73–Q75]
- Jaisaa ki oopara bataay gaya hai, ye highly worldsaneey dhruveey satellite launch vehicle (Polar Satellite Launch Vehicle) tha, specially PSLV-C37 mission, jisne 104 satelliteon ka meela ka patthar haasil kiya. Kailendar varsha 2022 ko banda karte hue, ISRO ka last peesaelavee (PSLV) launch PSLV-C54 mission tha. Isane 26 November 2022 ko Earth observation satellite-06 (oshanasaita-3) aur aatha nano-satelliteon ko successfully orbit me establish kiya. Nyooospe indiya limited (NSIL) ke through nishpaadit ek dedicated commercial mission me, PSLV-C55 rocket ne 22 aprail 2023 ko do singaapur ke satelliteon, TeLEOS-2 aur Lumelite-4 ko successfully orbit me establish kiya. [Q76–Q78]
- 1 janavaree, 2024 ko poori tarah se launch kiya gaya, XPoSSat black hole aur neutron sistars jaisee ghatanaaon ka study karne ke liye dedicated space-aadhaarita eksa-re polaarimetree me India ka pioneer pravesh hai (statement 1). SITE (saitelaaita instrakshanal television eksaperimenta) ko 1975 me launch kiya gaya tha. APPLE (erivehicle paisenjar peloda eksaperimenta) satellite ko 19 June 1981 ko French Guana ke Kourou se eriyen launch vaahan dwara launch kiya gaya tha. [Q79–Q81]
- Jaisaa ki pehle bataay gaya hai, India ka pehla praayogik geostationary Communication Satellite, APPLE, June 19, 1981 ko launch kiya gaya tha. INSAT-1D ko June 12th, 1990 ko launch kiya gaya tha. Ye ek bahukriyasheel (multifunctional) satellite tha jo India ke liye telephone, television aur weather observation service provide karta tha. INSAT-2A India ka pehla indigenously built satellite hone ka gaurav rakhta hai. Ise July 10th, 1992 ko French Guana ke Kourou se successfully space me launch kiya gaya tha. [Q82–Q84]
- 10 July 2006 ko INSAT-4C Communication Satellite le go rahaa GSLV (Geosynchronous Satellite Launch Vehicle) apne maarga se bhatak gaya aur liphta-opha ke immediate baad usamen visphot ho gaya. Usa time INSAT-4C ka weight 2168 kggraam tha. IRNSS-1A Indian kshetreey nevigeshan satellite system (IRNSS) ke hisse ke roop me launch kiya gaya pehla satellite tha. IRNSS starmandal (jisakaa naam NavIC hai) me saata active satellite shamil hain: teen geostationary orbiton me aur chaara jhukee hui bhoosamakaalik orbiton me located hain. Isliye, statement 1 sahi hai. [Q85–Q87]

- NavIC (naabhika) Indian kshetreey nevigeshan satellite system (IRNSS) ko diya gaya parichalan naam hai. Is short naam ka meaning "Navigation with Indian Constellation" hai. Listed options me se, Japan ka apana freelance kshetreey satellite nevigeshan system hai jise kvaasee-jenitha satellite system (QZSS) kaha jata hai, jise michibikee (Michibiki) bhi kaha jata hai. Galileo projekt (Galileo Project) European Union ka freelance vaishvik satellite nevigeshan system hai. Ise European space Agency (ESA) ke through banaya gaya tha taaki American GPS aur Russian GLONASS systems se freelance ek vaikalpik aur highly accurate sthiti service provide ki go sake. [Q88–Q90]
- GPS ka matalab Global Positioning System hai. Ye joint raajya USA ki government ke svaamitv wala aur yoonaaited stetsa space phorsa dwara operated widely use kiya jane wala space-aadhaarita rediyo nevigeshan system hai. Global Positioning System (GPS) sirpha nevigeshan se kahin zyada kaam karta hai. Ye aworldsaneey roop se accurate time signal (timing signal) provide karta hai. India ke launch yanon ka development maamoolee shuruaat se hua aur ye lagaataar saksham hota gaya: [Q91–Q93]
- (i) SLV-3 ne 1980 me Rohini satellite (RS-1) ko orbit me successfully establish kiya tha (B se mela khaata hai). [Q94]

6. ISRO Centres, Sriharikota, Thumba & Aviation Basics

- India me ina instituteon ke bhaugolik jnyaan ke basis par: Hassan me maastar kontrol phaisilitee (MCF) Karnataka raajya me ISRO dwara established ek important establishment hai. A. ISRO (Indian space research sangathana) ka mukhyalay Bengaluru (3) me located hai. [Q95–Q97]
- ISRO satellite sentar (jisakaa naam badalakar 2018 me yoo aara Rao satellite sentara, URSC kar diya gaya) satelliteon ke development ke liye major hub hai aur ye Bengaluru, Karnataka me located hai, na ki Sriharikota me. Sriharikota sateesh dhavan space hub (praathamika launch suvidhaa) ka ghara hai. Space application hub (SAC) Ahmedabad, Gujarat me apne sthana ke saath correctly mela khaata hai. A. VSSC ka milaan trivendram (Thiruvananthapuram) se correctly kiya gaya hai. [Q98–Q100]
- Indian sudoor sanvedan institute (IIRS) ISRO ke under ek major prashikshan aur academic institute hai. Ye uttaraakhand ke Dehradun me located hai. Thiruvananthapuram me Vikram Sarabhai space hub (VSSC) ISRO ka major hub hai jo launch vehicle (launch vehicle) technology ke design aur development ke liye jimmedaar hai. Ye mainly rocket launch karne ke liye pranodak (propellants) me research aur satellite launch vehicle (jaise PSLV, GSLV) ke samagr development se related hai. PraVaHa ISRO ke Vikram Sarabhai space hub (VSSC) me developed ek lesspyooteshanal phlooid daayanemiks (CFD) software hai. Isliye, statement 1 sahi hai. [Q101–Q103]
- Thumba ikvetoriyal rocket lonching Station (TERLS) 1962 me established ek Indian spesaport hai. Ye Thiruvananthapuram, Kerala me located hai, jo main ground India ke dakshinee sire ke paas, Earth ke magnetic bhoomadhy line ke bahut kareeb hai. Sriharikota Andhra Pradesh ke nellor jile me shaara (SHAR) project bastee me located bangaal ki khaadee ke tata se doora ek baadha dveepa hai. Ye Pulicat lake (Pulicat Lake) ko bangaal ki khaadee se alaga karta hai. Sriharikota, jo sateesh dhavan space hub ka ghara hai, Andhra Pradesh raajya me located hai. [Q104–Q106]
- Kaakaraapaar ek nuclear energy sandevice ka sthala hai (A-3). 23 pharavaree, 2004 ko, Sriharikota me sateesh dhavan space hub ke andara located tosha pranodak (solid propellant) sandevice me ek bhayankar visphot aur uske baad aaga laga gayi thi, jiske parinaamasvaroop hatahat hue the aur suvidha ko damage pahuncha tha. Havaaee jahaaj ke pankhon ki ooparee surface uttala (baahara ki direction ghumaavadaara) hoti hai aur nichalee surface chaapaloossee/avatala (eyaraphoila aakaara) hoti hai taaki air flow me herapher kiya go sake. Isliye abhistatement (A) true hai. [Q107–Q109]
- naam ke baavajooda, ek havaaee jahaaj ke 'blaika boksa' (phlaaita data rikordara) ko chamakeele, garmee pratirodhee "international naarangee" (Orange) ranga me rangaa jata hai taaki ise highly visiblenamean banaye go sake aur durghatana ke baad malabe me discoverya aasaan ho sake. [Q110]

7. Moon Missions: Chandrayaan, LUPLEX, Luna & Aditya-L1

- Moon Earth ka elessaatr praakritik satellite hai. Ye Earth ke charon direction orbit karta hai. Moon par India ka pehla mission Chandrayaan-1, 22 aktoobar 2008 ko launch kiya gaya tha. Kai yuddhaabhyason ke baad, vaahan ko 8 November 2008 ko lunar orbit me successfully establish kiya gaya tha. India ke dedicated lunar anveshan missionon ko diya gaya naam "Chandrayaan" hai. [Q111–Q113]
- Chandrayaan-1 ko Andhra Pradesh raajya ke Sriharikota dveepa par located sateesh dhavan space hub se launch kiya gaya tha. Chandrayaan-2 project ke early yojana aur developmentaatmak charanon ke dauran (jab ye question 2010 ke aasapaas taiyar kiya gaya tha), do mayalasvaamee annaaduraee (jinhone Chandrayaan-1 ka bhi netrity kiya tha) project ka netrity kar rahe the. Chandrayaan-2 mission ka main purpose anveshan ke liye lunar dakshinee dhruva (South Pole) field ko lakshit karna tha, na ki uttaree dhruva ko. Isliye statement (C) true nahi hai. [Q114–Q116]
- Chandrayaan-2 ko Sriharikota ke sateesh dhavan space hub se launch kiya gaya tha, jo Andhra Pradesh raajya me located hai. Chandrayaan-2 spacecraft ko orbit me launch karne ke liye heavy-bharaless (heavy-lift) lonchar GSLV-MkIII-M1 (jise aba aksar LVM3 ke roop me naamit kiya jata hai) ka successfully use ISRO dwara kiya gaya tha. Statement 1 accurate hai. Chandrayaan-2 laindar ki aanshik viphalata ke baad, Chandrayaan-3 ko ek pratyaksh anuvartee mission ke roop me design kiya gaya tha jo specially ek successful, safe sophta lainding mila karne aur lunar surface par rover ko tainaat karne par kendrit tha. [Q117–Q119]
- APXS (alphaa paartikal eksa-re spektromeetara) Chandrayaan-3 mission ke prajnyaan rover par lagaa ek specific scientist device tha. Loonar polara eksaploreshan (LUPLEX) mission India ke ISRO (ISRO) aur Japan ke jaaksa (JAXA) ke beech ek sahayogaatmak prayas hai jo Moon ke dakshinee dhruva ki jaanch par kendrit hai. (statement 1 sahi hai). Moon Express Moon par jane wali world ki pehli private flight yojana hai. 2017 se kaaryanvit is yojana ko 3 agasta, 2016 ko American sangheey uddayan prashaasan dwara sahamati provide ki gayi thi. [Q120–Q122]

- June 2024 me, China ka chaanga'ee 6 (Chang'e 6) mission Moon ke sudoor hisse (andhere hisse) se Earth par mittee laane wala pehla lunar mission bana gaya. Loonaa-25 agasta 2023 me ek asuccessful Russian lunar laindar mission tha. Iski yojana international lunar research Station (ILRS) ke hisse ke roop me banayi gayi thi. Aditya-L1 Sun ka study karne wali pehli Indian space-aadhaarita observatory hai. Ise 2 sitanbar 2023 ko Shriharikota se launch kiya gaya tha. [Q123–Q125]
- Aaditya-L1 me Photosphere (photosphere), kromospheeyar aur korona ka nireekshan karne ke liye saata scientist peloda hain. Chaara peloda directly Sun ko dekhate hain (rimota sensinga) aur shesha teen L1 point par particles aur kshetron ka ina-seetoo observation study karte hain. [Q126]

⚠ EXAM TRAP — Q113 — Multiple Correct Answers

► Source Q113 ko multiple-correct mark karta hai. Chandrayaan-I se related naming/mission statements ko exact option wording ke context me evaluate karein.

8. Mangalyaan, China & International Space Cooperation

- Maarsa orbitar mission (MOM), jise anaupacharik roop se Marsvehicle ke naam se jana jata hai, India ka pehla antaraplaneteey mission tha. (statement I sahi hai). India ka pehla intaraplenetaree mission, maarsa orbitar mission (Marsvehicle), 5 nananbar 2013 ko launch kiya gaya tha. Marsvehicle ko maarsa orbitar mission (MOM) bhi kaha jata hai. (statement 1 sahi hai). [Q127–Q129]
- Tiyangong Space Station China dwara built ek sthayee roop se crew wala Space Station hai. Iska core module 'Tianhe' hai, pehla Lab module 'Wentian' tha, aur aktoobar 2022 me launch kiya gaya doosra aur last module 'Mengtian' (Mengtian) tha. Tivehiclevena-1 China National space prashaasan (CNSA) ka ek antaraplaneteey mission hai jisme Mars planet par bheja gaya ek robotik spacecraft (spacecraft) shamil hai (jisme ek orbitara, laindar aur joorong rover shamil tha). China ne sitanbar 2008 me country ke pehle spesavok ka prayas karne ke liye *teen* (chaara nahi) space yatriyon ke saath shenjhou-7 launch kiya tha. Isliye, statement (A) sahi nahi hai. [Q130–Q132]
- Misiyas (Micius), jiska naam ek praaChina Chinaee daarSaturnka ke naam par rakha gaya hai, world ka pehla kvaantam Communication Satellite (QUESS) hai, jise agasta 2016 me China dwara launch kiya gaya tha. NISAR ka full form NASA-ISRO synthetic aparchar Radar (NASA-ISRO Synthetic Aperture Radar) hai. Ye NASA aur Indian space research organisation (ISRO) ke beech ek joint Earth-avalokana mission hai. CNES (neshanala sentar phora space stadeeja) France ki government space Agency hai. [Q133–Q135]
- Russia ne Arctic field me climate aur paryavaran ki nigaraanee ke mission par pharavaree 2021 me apanaa space satellite Arcticaa-ema (Arktika-M) launch kiya. Navanbar 2022 me, Japan ki space Agency JAXA ne ghoshana ki ki usane apne water-sanchaalita kyoobasait EQUULEUS ko aage badhane ke liye successfully steam ka use kiya hai. Water pranodak system use karke kam-Earth ki orbit se pare ye world ka pehla successful orbit control tha. Pharavaree 2018 me, JAXA (Japan eyarospes eksaploreshan Agency) ne ek maaikrosaitelaaat ko orbit me established karne ke liye sanshodhit ek SS-520 saaunding rocket ko successfully launch kiya, jisse ye satellite ko orbit me le jane wala aba tak ka sabse small rocket bana gaya. [Q136–Q138]
- NASA (NASA) ki jeta propalshan leboetaree (JPL), jo Agency ke kai robotik space missionon ka prabandhan karti hai, joint raajya USA ke California raajya ke Pasadena me located hai. Greed laaitaning ya jeela-10 prototype NASA dwara test kiya gaya baitaree se chalne wala 10-engine wala electric aircraft hai. Ise heleekoptar ki tarah lanbavat flight bharane aur utarane ke liye design kiya gaya hai, lekin pankhon wale aircraft ki tarah kushalata se udataa hai. [Q139–Q140]

9. NASA Missions, Planets, Jet/Rocket Principles & Cryogenics

- NASA ne aktoobar 2020 me pushti ki ki uske mission, SOFIA (straitospherika objarvetaree phora inphraared estronomee) ne kleviyas kretar me Moon ki Sun ki roshanee wali surface par water ke anuon ki upasthiti ki pushti ki hai. Comet ke internal bhaaga ki structure ka study karne ke liye NASA ka deepa impaikt mission (Deep Impact Mission) launch kiya gaya tha. Is spacecraft ne 4 July 2005 ko kometa tempel 1 (Temple-1) me ek impaiktar ko successfully kraish (takkara) kiya tha. Die gaye options me, Phoenix NASA ke maarsa skaaut programme ke under Mars planet par space anveshan mission par ek robotik spacecraft tha. Ye 2008 me Mars planet par utara tha. [Q141–Q143]
- Spirit (MER-A), NASA ke Mars anveshan rover mission ke do judavaan rovars me se pehla tha. Ye 4 janavaree 2004 ko Mars planet par successfully utara. Iska judavaan, aporchunitee (Opportunity), kuch saptaah baad utara. NASA ke missionon ko Mars par ateeta ki water speedvidhi ke saakshya, jipsam ke bhandaar (water ki upasthiti me banane wala ek khanija), aur Methane gas ke gubaar (jo Earth par aksar jeevit jeevon dwara utpann hote hain) mile hain. Isliye, ye teenon saamoohik roop se Mars planet par ateeta ya vartamaan jeevan ki sanbhaavana badhaate hain. July 2018 me, shodhakartaaon ne Radar data ka use karte hue Mars planet par tarala water ki ek huge bhoomigat jheela ka detect lagaane ki information dee, jo planet ke dakshinee dhruveey barpha ki topee ke neeche lagbhag 1.5 km neeche located hai. [Q144–Q146]
- Vibhinn orbitars aur rovars dwara Mars ke vaayumandal me Methane ka detect lagaay gaya hai. Iske alawa, Jupiter ke saath-saath Saturn, yoorenas aur nepachyoon ke vaayumandal me naturally Methane ki amount paaee jati hai. Isliye, Mars aur Jupiter donon sahi option hain. Saturn ka vaayumandal mainly haaidrojan aur hiliyam se bana hai, lekin die gaye options me, Carbon monoxide, Nitrogen ya Sulphur dioxide ke comparison me Methane sabse important amount me maujood hai. Hamare saura mandal ke sabhi planeton me Saturn ka ausata ghanatv sabse kam hai, lagbhag 0.7 graama prati ghana senteemeetara. Iska ghanatv water se bhi kam hai, jiska meaning hai ki ye saiddhaantik roop se water ke kaaphee bade poola me taira sakta hai. [Q147–Q149]
- NASA (NASA) ke spacecraft (jaise Discovery) ke external tainka me Liquid hydrogen fuel aur Liquid oxygen okseedaaijar bharaa hota hai jo liphta-opha ke dauran shuttle ke main engineon ko power provide karta hai. Moola pareeksha question me diya gaya koi bhi option sahi nahi hai. Apolo 8 (Apollo 8) pehli maanavayukt space flight thi jisne Earth ki nichalee orbit ko chhoda Moon tak reach aur usakee orbit karke lautane ka work kiya tha. Ise 21 disanbara, 1968 ko launch kiya gaya tha.

SELENE (selenologikala enda engineering eksplorara), jise iske upanaam 'kaaguya' se bhi jana jata hai, doosra Japanee (Japan) lunar kaksheey spacecraft tha. Ise JAXA dwara 2007 me launch kiya gaya tha. [Q150–Q152]

- Ek jeta engine ko apanaa fuel jalaane ke liye essential okseejan mila karne ke liye Earth ke vaayumandal ki requirement hoti hai, isliye ye space ke nirvaat me kaam nahi kar sakta hai (statement 1 sahi hai). Commercial jeta aircraft vaayumandaleey vikshobh (turbulence) se bachane ke liye aksar nichale Stratosphere me flight bharate hain. Choonki isme lagbhag koi water vaashp nahi hota hai, isliye weather-sanbandhita vikshobh paidaa karne ke liye practical roop se koi baadal nahi hote hain. Ek jeta engine raikhik sanveg (linear momentum) ke sanrakshan ke principle par kaam karta hai, jo anivaary roop se nyootan ke speed ke teesare niyama (pratyeka kriya ke liye, ek same aur vipareet pratikriya hoti hai) ka ek application hai. Peechhe ki direction chhodde gayi high speed wali nikaas gaisen aage ki direction dhakka (thrust) paidaa karti hain. [Q153–Q155]
- Jeta engine ki tarah hi, rocket bhi sanveg sanrakshan (conservation of momentum) ke principle par kaam karta hai. Choonki garma nikaas gaisen peeche se teevra speed se dhakelee jati hain, rocket aage ki direction dhakela jata hai. 'straitalaaita' (Stratosphere aur satellite se bana word) ek human rahita, high elevation wala eyaraship (havaaee pota) hai jo Stratosphere me tainaat hota hai. Ye vaayarales communication ya observation provide karne ke liye satellite ke same kaam karta hai, lekin Earth ki surface ke bahut kareeb rahata hai. Physics me, kraayojeniks me atyant kam temperature par saamagriyon ke production aur behavior ka study shamil hai. Aama taur par, -150°C (ya 123 K) se kam ke kisi bhi temperature ko kraayojenik maana jata hai. [Q156–Q158]
- Kraayojeniks (bahut kam temperature ka vijnyaana) ka space yatraa (tarala fuel jaise haaidrojan aur okseejana), magnetic uttalan (maigleva trenon me use kiye jane wale suparakandakting ilektromagnets ko thanda karna), aur sensitive teleemetree device (riseevra me tharmal shora ko kam karne ke liye) me important application hai. Kraayojenik engine vishesh rocket engine hain jo kraayojenik fuel ya okseedaaijar ka use karte hain, jiska meaning hai ki gaison ko taraleekrit (liquefied) kiya gaya hai aur atyant kam temperature par sanplaneteet kiya gaya hai. Ve rocket (jaise jeeesaelavee) ke ooparee charanon ke liye highly kushal pranodan provide karte hain. Kraayojenik engine aama taur par okseedaaijar ke roop me Liquid oxygen aur fuel ke roop me Liquid hydrogen ka use karta hai. Inhen tarala state me banaye rakhne ke liye atyant kam temperature par storage ki requirement hoti hai. Space shuttle kraayojenik engine ka bhi use kiya jata hai. [Q159–Q161]
- Behada thande tarala fuel use karke high thrust (dhakkaa) provide karne ke liye Space Shuttle aur advanced rocket launch vaahanon (jaise GSLV) me kraayojenik engine ka use kiya jata hai. Die gaye options me se sabhi ka use rocket pranodak (propellant) ke roop me kiya jata hai. Liquid hydrogen, Liquid hydrazine aur mittee ke tela (RP-1) ka use fuel ke roop me kiya jata hai jabki Liquid oxygen ka use okseedaaijar ke roop me kiya jata hai. Pranodak fuel aur okseedaaijar ka mishran hai. [Q162–Q163]

⚠ EXAM TRAP — Q147, Q151 & Q163 — Source-Sensitive Questions

- ▶ Q147 source me multiple-correct hai: methane ek se adhik planetary atmospheres/context me mil sakti hai.
- ▶ Q151 ke options/date wording me source error/dispute marked hai; Apollo-8 historical date ko standard reference se cross-check karein.
- ▶ Q163 rocket propellant question source me disputed hai; propellant classification fuel + oxidizer pair par depend karti hai.

10. Indian Launchers, Satellites, EDUSAT, GSAT & BHUVAN

- India electronics limited (BEL) GaAs (gailiyama aarArmyida) saura sela developed karne ke liye ISRO ke saath sahayog kar rahaa hai. Space ke liye ye saura sela technology ke maamale me zyada advanced hain aur space me bijalee paidaa karne me kushal hain, jo Indian satelliteon ki deerghaayu me help karte hain. India ne aktoobar 2006 me 50 sekand ke liye ek full kraayojenik charan ka successfully test kiya tha (statement 1 sahi hai). RISAT-2 ek Indian Radar tohi (nigaraanee) satellite hai jise seemaan ki nigaraanee ke liye design kiya gaya hai. Ise 20 aprail 2009 ko PSLV-C12 rocket dwara successfully launch kiya gaya tha. [Q164–Q166]
- Statement 1 incorrect hai kyonki INSAT series ke sabhi satelliteon ko videshon se launch nahi kiya gaya tha. Example ke liye, INSAT-4C ko 10 July 2006 ko Shriharikota se launch kiya gaya tha. Statement 1 sahi hai: PSLV ka use mainly Earth-avalokana (rimota sensing) satelliteon ko dhruveey orbiton me established karne ke liye kiya jata hai, jabki GSLV ko heavy Communication Satelliteon ko high bhoosamakaalik orbiton me established karne ke liye design kiya gaya tha. Doppler effect use karke ek kritrim satellite ko Earth se bahut accurate roop se traika kiya ja sakta hai. Satellite lagaataar rediyo signal emit karta hai, aur graaund Station ina sanketon ki frequency (dopalara shiphta) me badalaav ka analysis karte hain kyonki satellite unki direction ya doora jata hai, jisse iski accurate speed aur prakshpevakr (trajectory) ki calculation ki jati hai. [Q167–Q169]
- SLV-3 (saitelaaita launch vehicle-3) koi space satellite nahi hai; ye satelliteon ko orbit me established karne ke liye ISRO dwara developed ek rocket/launch vaahan hai. GSLV ek satellite launch vikal (rocket) hai, ye apne aapa me koi satellite nahi hai. GSLV-D2 is vaahan ki doosri developmentaatmak test flight thi (maee 2003 me launch ki gayi). Statement 1 incorrect hai: science rela rela ministry aur 'vijnyaana prasaara' (vijnyaana aur technology department ke under) dwara joint roop se organized kiya gaya tha, na ki mainly CSIR dwara. [Q170–Q172]
- EDUSAT (GSAT-3) specially shiksha field ki maangon ko poora karne ke liye dedicated satellite hai. Ise GSLV-F01 rocket use karke 20 sitanbar 2004 ko Shriharikota se launch kiya gaya tha. India ka pehla full Meteorological Satellite 'METSAT' 12 sitanbar 2002 ko successfully launch kiya gaya tha. (statement 1 sahi hai). 2002 me launch kiye gaye India ke pehle dedicated Meteorological Satellite METSAT ka naam baad me pradhaan mantree atala Biharee vaajapeyee dwara Indian moola ki divangat NASA astronaut kalpana chaavala (jinakee Columbia Space Shuttle durghatana me mrityu ho gayi thi) ke sammaan me kalpanaa-I rakha gaya. [Q173–Q175]
- Jaisaa ki pichhale questionon me shamil kiya gaya hai, Meteorological Satellite METSAT (baad me KALPANA-1) ko ISRO dwara sitanbar 2002 me launch kiya gaya tha. SATYABAMASAT (Sathyabamasat) ek microscopic praayogik satellite hai

jise chennnew ke satyabhaama worldvidyalay ke chhaatron aur shikshakon dwara greenahaaus gaison par data ekatra karne ke liye develop kiya gaya hai. Isliye, ye ek academic institute ka satellite hai. Apne heavy weight ki wajah se, India ka advanced Communication Satellite GSAT-8 21 mae 2011 ko dakshin USA ke French Guana me Kourou se erivehicle-V launch vehicle dwara successfully launch kiya gaya tha. [Q176–Q178]

- GSAT-7 (jise rukminee ke naam se bhi jana jata hai) ISRO dwara developed ek advanced maltee-band Communication Satellite hai. Ye specially Military communication ke liye Indian Navy ko dedicated hai, jo iski bloo-vaatara capacityon ka expansion karta hai. GSLV-D3 rocket ke teesare charan ki viphalata ki wajah se GSAT-4 (April 2010 me launch kiya gaya) orbit tak reach me viphal rahaa. 29 sitanbar 2012 ko India ne guvehicle space hub se erivehicle-5 rocket use karke heavy dooraCommunication Satellite GSAT-10 ko successfully launch kiya tha. [Q179–Q181]
- GSAT-6A, ek high power wala esa-band Communication Satellite, ISRO dwara 29 maarch 2018 ko Shriharikota se GSLV-F08 rocket use karke launch kiya gaya tha. (nota: launch ke kuch time baad hi satellite se sanpark toota gaya tha). Bhuvan (Bhuvan) ISRO dwara developed ek web-aadhaarita upayogita (sophtaveyara application/jiyportal) hai. Ye users ko Earth ki surface ke 2dee/3dee represent ka detect lagaane ki allow karta hai, jo Google meaning ke Indian version ke same kaam karta hai. [Q182–Q183]

✓ Source coverage: Q1–Q183